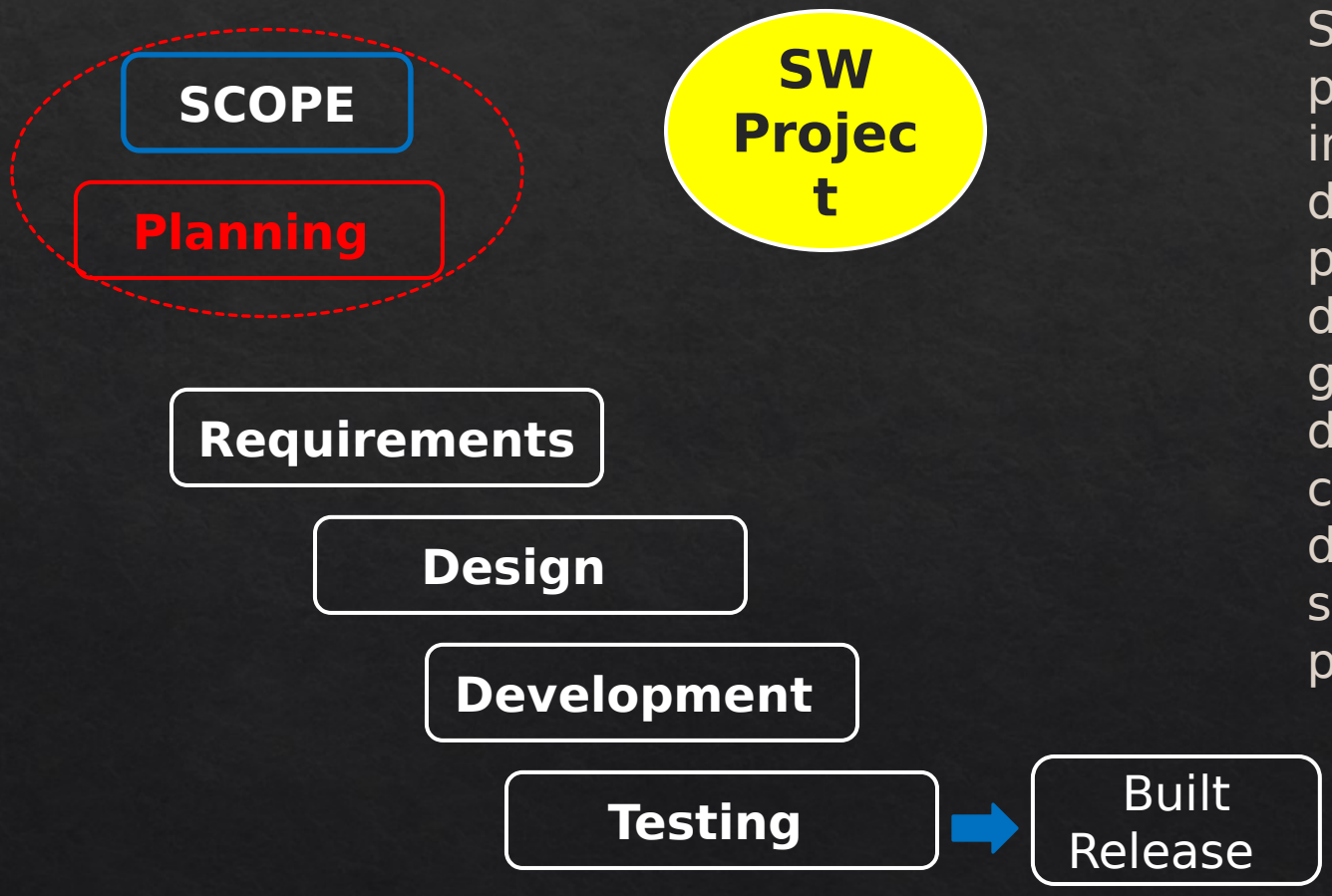
Work Breakdown Structure



Scope and planning involves defining a project's deliverables, goals, tasks, deadlines, and cost and how to develop the software product.

Project Management

Planning

Define activities & deliverables

Estimating

Time, Cost in terms of resource usage

Scheduling

Define sequence of tasks occurrence

What is WBS?

a work breakdown structure is:

- “A hierarchical decomposition
- of the total scope of work
- to be carried out by the project team
- to accomplish the project objectives
- and create the required deliverables.”

WBS

Dividing complex projects to simpler and manageable tasks.

Stops till the tasks no longer remains decomposable.

Work packages: the smallest unit of work that a project can be broken down into.

A WBS element may be a product, data, a service, or any combination.

Why is it needed?

- For activities identification
- For correct estimation
- Identify related activities using PERT
- Creating schedules
- Project failure risk minimized

When should we develop WBS?

- **After finalizing project scope and before project estimation**

How to create WBS?

Done by project managers mostly

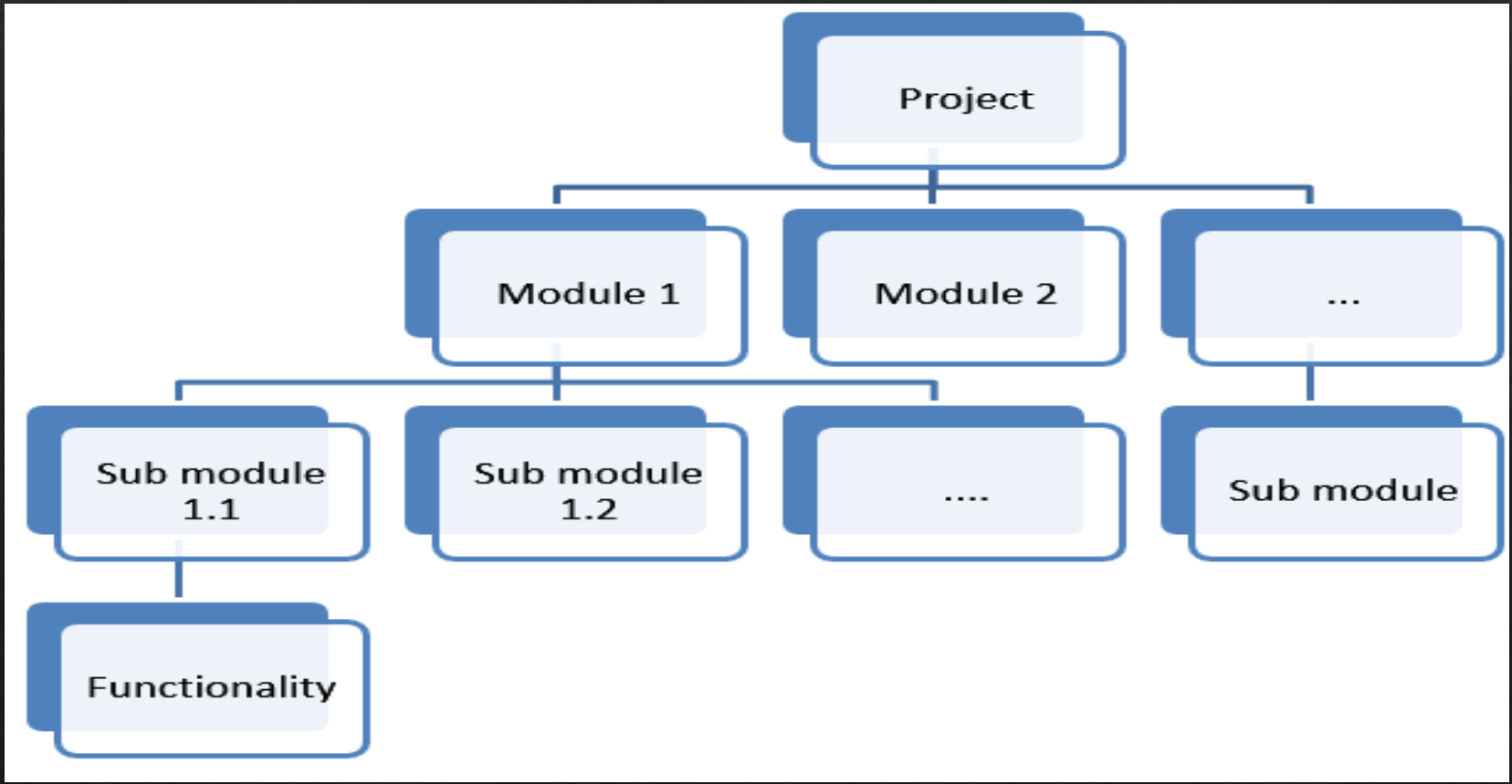
describe the project statement

Highlight all necessary phases in the project

Create deliverables

Validate for completeness

WBS


Project
Module 1
Module 2
...
Sub module 1.1
Sub module 1.2
....
Sub module
Functionality

Reasons to use WBS

correct project organization.

depicts whole project work.

Helpful for describing the project scope to stakeholders.

Helpful in cost estimation, resource allocation, and time management improvements.

Minimize unnecessary work that leads to missed deadlines and out-of-scope work.

Levels in WBS

- **Starts with level 0 i.e. project title**
- **Then make level 1, highest level breakdown of about 4-7 nodes depending on your project. Termed as control accounts**
- **Then make level 2, they are work packages.**
- **Level 3 are the tasks/activities to be performed**
- **Follow the standard numbering scheme**

WBS Formats

OUTLINE (INDENTED FORMAT)

GRAPHICAL TREE (ORGANIZATIONAL CHART)

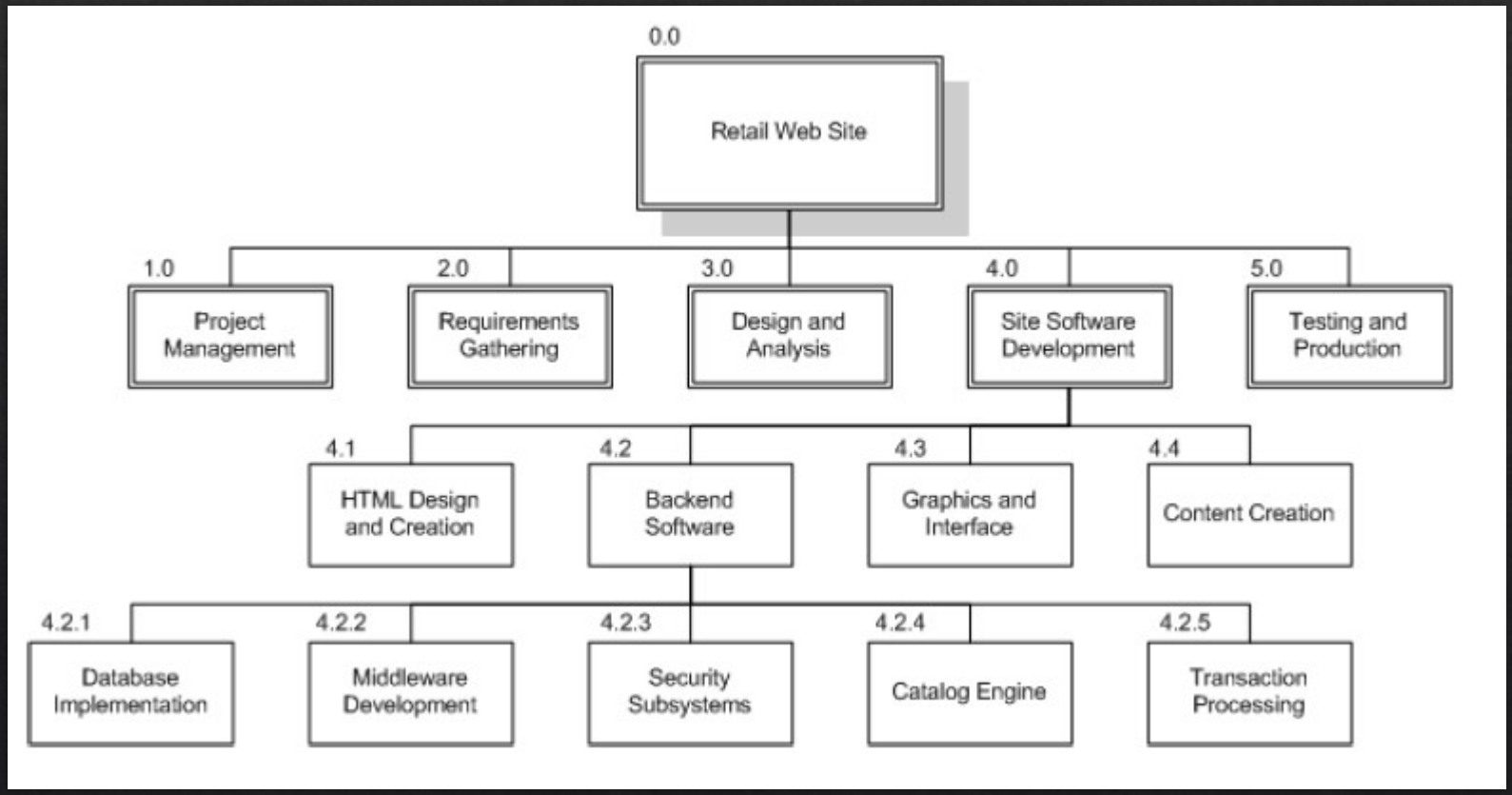
Displaying the WBS

Example of outlined WBS.

- 0.0 Retail Web Site
- 1.0 Project Management
- 2.0 Requirements Gathering
- 3.0 Analysis & Design
- 4.0 Site Software Development
  - 4.1 HTML Design and Creation
  - 4.2 Backend Software
    - 4.2.1 Database Implementation
    - 4.2.2 Middleware Development
    - 4.2.3 Security Subsystems
    - 4.2.4 Catalog Engine
    - 4.2.5 Transaction Processing
  - 4.3 Graphics and Interface
  - 4.4 Content Creation
- 5.0 Testing and Production

Displaying the WBS

Example of Graphical Tree WBS



Challenges

Must have concrete knowledge of project

All tasks/ deliverables properly identified and quoted

Each task should be broken down into legit activities

WBS Guidelines

- MUST BE **EASY TO UNDERSTAND**
- COVERS ALL OBJECTIVES
- EACH ELEMENT CORRESPONDS TO A DELIVERABLE
- BREAKDOWN UNTIL YOU GET ACCURATE TIME AND COST ESTIMATES
- SHOULD NOT BE MUCH DETAILED AS PROJECT PLAN
- CAN EVOLVE OVER TIME
- EACH LEVEL HAVE NO MORE THAN 7 ITEMS
- CAN BE DRAWN USING GANTT PRO, WORD, EXCEL, MS PROJECT, VISIO ETC. MAY USE A TEMPLATE IF YOU HAVE

Example WBS

WBS

- **Redecorate Room**
  - **Prepare materials**
    - **Buy paint**
    - **Buy a ladder**
    - **Buy brushes/rollers**
    - **Buy wallpaper remover**
  - **Prepare room**
    - **Remove old wallpaper**
    - **Remove detachable decorations**
    - **Cover floor with old newspapers**
    - **Cover electrical outlets/switches with tape**
    - **Cover furniture with sheets**
  - **Paint the room**
  - **Clean up the room**
    - **Dispose or store left over paint**
    - **Clean brushes/rollers**
    - **Dispose of old newspapers**
    - **Remove covers**

Professional Tools used

- https://www.projectmanager.com/
- https://www.atlassian.com/software/jira
- https://ganttpro.com/